

Joshua M. Hofer

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'DEFENSE MANUFACTURING | ELECTROMECHANICAL DESIGN | TEST SYSTEMS

EDUCATION

Wentworth Institute of Technology

Bachelor of Science in Electromechanical Engineering, Manufacturing Minor

Boston, MA

Expected August 2026

WORK EXPERIENCE

Dynalec Corporation – Electromechanical Engineering Co-op

Sodus, NY | May 2025 – January 2026

- Designed and built electromechanical fixtures used to test and assemble Navy shipboard components, including course indicator and airflow indicator systems.
- Created interconnect diagrams, PCB trace layouts, and 3D CAD models for a synchro test fixture, improving operator workflow, testing repeatability, and manufacturability.
- Collaborated with technicians and operators to troubleshoot production and RMA test issues and implement practical fixture and process improvements.
- Supported manufacturing in a defense-focused environment by improving test efficiency, fixture reliability, and technical documentation.

Consolidated Precision Products (Wollaston Alloys Inc.) – Foundry Engineer Co-op

Braintree, MA | May 2024 – January 2025

- Supported precision casting and process engineering work relevant to defense manufacturing in an ITAR controlled environment.
- Led a hot topping and riser efficiency study focused on improving casting yield and reducing defects.
- Used MAGMASOFT simulation software to analyze mold filling, solidification, and defect formation in castings.
- Performed RFID sensor testing and calibration while supporting industrial automation and lean manufacturing initiatives.

Research Assistant to Dr. Serdar Tumkor, Ph.D.

Boston, MA | January 2026 – Present

- Supported research related to hybrid aluminum-steel armor architecture through technical investigation, materials evaluation, and research documentation.

Accelerate– Innovation & Entrepreneur Collaboration Space – Collaboration Support

Boston, MA | September 2022 – November 2024

- Supported entrepreneur product development through CAD, 3D printing, laser engraving, woodwork, and rapid prototyping.

PROJECTS

Cast in Steel Competition / TV-Series – Team Lead

Nov 2023 – Present

- Led student engineers through the design and production of DoD-sponsored competition castings, overseeing CAD, simulation, foundry coordination, testing, and documentation for the Halligan Bar and George Washington cuttue.

Automated Vertical Farm

2026

- Developing an automated indoor growing system integrating sensing, environmental monitoring and control, AI vision learning, embedded systems, and robotic arm integration for plant monitoring and automated maintenance.

Custom FPGA Tetris Game

Spring 2024

- Developed and programmed a Tetris game in Verilog on a DE10 Lite FPGA board.

LEADERSHIP & ACTIVITIES

Boston College Marching Band and Pep Band

Musician

Boston, MA

Aug 2022 – December 2024

American Foundry Society

Club President

Boston, MA

2023 – 2024

Boston Civil Council, Worship Team, Tutoring Center, IEEE, Bald For Bucks, Class President

CORE COMPETENCIES

Software: SolidWorks, Onshape, Autodesk Suite, MAGMASOFT, Microsoft Office, Access, Power Automate, AI Tools, Visual Studio, C++, Verilog, MATLAB, Multisim

Technical: Test Fixture Design, Electromechanical Assembly, GD&T, Technical Drawing, PCB Layout, Embedded Systems, Microcontrollers, Sensor Integration, Welding, Metal Fabrication, Sand Casting, Schematic Interpretation, Process Improvement

Leadership: Loving Leadership, Team Coordination, Organization, Cross-Departmental Collaboration

Certifications: SOLIDWORKS Associate, Forklift Certification

Awards: “Wilkes Desmond Scholarship”, “Nuts, Bolts, Thingamajigs Scholarship”